Rushabh Shah

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Summary

Senior Software Engineer with 10+ years of experience designing scalable, cloud-native distributed systems. Architected event-driven pipelines processing **60M+ events/day** on AWS, led cross-functional platform initiatives, and delivered developer tooling that measurably reduced cycle times. Deep expertise in serverless architecture, microservices, and full-stack development with a strong track record of owning high-impact, business-critical platforms end-to-end. **Adaptable and eager to learn new technologies, with a strong ability to quickly acquire new skills and apply them effectively to solve complex engineering problems.**

Certifications

AWS Certified Solutions Architect – Professional

AWS Certified Solutions Architect – Associate

AWS Certified Developer – Associate

AWS Certified AI Practitioner

Technical Expertise

Languages & Backend: TypeScript, JavaScript, Node.js, Express.js, SQL

Cloud & Architecture: AWS (Kinesis, Firehose, Lambda, Step Functions, EventBridge, S3, CloudWatch, CDK, CloudFormation), AWS Serverless Architecture, Microservices, Distributed Systems, System Design

Frontend: React, Angular, Micro-Frontends, Progressive Web Apps (PWA)

Databases: DynamoDB, MySQL, MongoDB, PostgreSQL (prior experience)

Testing & APIs: Jest, REST APIs

DevOps & Tools: Docker, Infrastructure as Code, CI/CD, Azure AD (SAML), Datadog, New Relic

Practices: Technical Design & Leadership, Code Review, P0 Incident Response, Observability, RBAC, Security Engineering

Professional Experience

PowerSchool - Senior Software Engineer II

Nov 2021 – Present
Bangalore, India

Project: Usage Analytics Platform

Skills: AWS Kinesis Streams, Kinesis Firehose, Node.js, TypeScript, AWS Serverless, Event-Driven Architecture, S3, CloudWatch, AWS CDK, CloudFormation

Contributions & Business Impact: Architected and led the migration of a manually managed analytics pipeline to AWS Kinesis Streams and Firehose, building a resilient, serverless platform that processes **60M+ events/day** while significantly reducing operational overhead.

Designed the platform for high-volume event ingestion and processing, improving scalability, reliability, and operational efficiency.

Implemented event-driven processing patterns to decouple services and improve system resilience.

Provisioned infrastructure using **AWS CDK/CloudFormation**, enabling repeatable infrastructure deployment and reducing manual infrastructure management.

Project: Student & Staff Data Integration Platform

Skills: AWS Lambda, Step Functions, EventBridge, S3, CloudWatch, AWS CDK, CloudFormation, Node.js, TypeScript, ETL, Serverless Architecture

Contributions & Business Impact: Designed and developed a cloud-native ETL platform to ingest Student and Staff data from SIS platforms, enabling reliable large-scale data integration across enterprise customers.

Designed serverless workflows using **Lambda, EventBridge, Step Functions, and S3** to orchestrate asynchronous data processing.

Implemented infrastructure using **AWS CDK and CloudFormation**, enabling repeatable and automated deployments.

Project: Core Product & Full-Stack Platform

Skills: Node.js, TypeScript, React, REST APIs, Jest, AWS, Micro-Frontends, CI/CD, Docker, Internationalization

Contributions & Business Impact: Developed and maintained full-stack applications using **Node.js, TypeScript, React, and AWS** with automated unit and integration testing using Jest, taking features end-to-end from technical design and implementation through CI/CD, production deployment, monitoring, and support.

Designed and developed **RESTful APIs and backend services** using Node.js and TypeScript, integrating them with React-based applications and AWS services.

Engineered a reusable **micro-frontend architecture** enabling plug-and-play integration across applications and accelerating cross-team feature delivery.

Led technical design discussions and code reviews across teams, **mentoring engineers** and driving architectural decisions for scalable, maintainable solutions.

Led platform internationalization and localization initiatives by introducing Arabic language support, RTL UI rendering, Hijri calendar integration, locale-aware formatting, and reusable localization patterns enabling multilingual product expansion.

Project: Developer Productivity & Operational Platform

Skills: VS Code, Cursor, MCP, Node.js, TypeScript, Docker, CloudWatch, Datadog, New Relic, Observability, Developer Tooling

Contributions & Business Impact: Built and shipped an internal **VS Code/Cursor extension** enabling developers to run and debug containerized test cases with one click, reducing environment setup time by **50%** and improving developer productivity.

Built an **MCP-powered operational assistant** integrated with VS Code/Cursor to centralize on-call runbooks, team responsibilities, and troubleshooting workflows, improving incident response efficiency.

Containerized services with **Docker**, enabling a fully local development environment and reducing cloud dependency.

Owned **P0 incident response and production reliability**, leveraging CloudWatch, Datadog, and New Relic for observability and troubleshooting to rapidly diagnose critical failures, restore services, and minimize business disruption.

Wipro - Senior Software Engineer

Jul 2019 – Nov 2021
Pune, India

Project: Multi-Tenant SaaS & Business Process Platform

Skills: Node.js, Angular, MongoDB, Express.js, BPMN, Jest, Azure AD, SAML, Firebase, PWA

Contributions & Business Impact: Engineered a multi-tenant SaaS platform using **Node.js, Angular, MongoDB, and Express.js**, delivering modules for Quality Checks, SLA management, notifications, and queue orchestration.

Developed a **BPMN-based process visualization module** enabling users to graphically design and configure business workflows for process digitization.

Developed modules for configuring and executing **Quality Checks and SLA rules**, supporting downstream reporting and operational workflows.

Developed configurable **notification and status-email workflows** triggered by business events, enabling timely communication of process and instrument status updates.

Designed a **Node.js security framework** with request sanitization and file-type verification based on file content; implemented automated unit testing with **Jest** and engineered centralized error handling to prevent stack trace exposure.

Integrated **Azure Active Directory using SAML-based SSO** for enterprise authentication and implemented end-to-end encryption in compliance with IRMC security standards.

Modernized the frontend into a **Progressive Web App (PWA)** with responsive UI across devices and integrated Firebase push notifications for real-time operational alerting.

Implemented **PDF document-splitting** functionality and structured logging to support automated document processing, observability, and analytics.

Infosys - Specialist Programmer

Dec 2016 – Jun 2019
Pune, India

Project: Multi-Tenant Health Insurance Platform

Skills: Node.js, Angular, MongoDB, Express.js, JavaScript, RBAC, Micro-UI

Contributions & Business Impact: Developed a **multi-tenant health insurance platform** for configuring insurance products and customer-specific applications, supporting flexible product setup and deployment.

Built a **dynamic form-generation engine** that rendered UI components from backend JSON configurations, supporting configurable layouts and validations and accelerating feature rollout.

Developed an **administrative portal** for application deployment and user management, enabling administrators to create applications and provision users with application-specific roles and permissions.

Implemented **Role-Based Access Control (RBAC)** to manage application-specific roles and permissions across multi-tenant applications.

Engineered reusable **micro-UI components** that could be injected into other applications, promoting modular architecture and cross-application reuse.

Education

Bachelor of Engineering, Information Technology 2016
Gujarat Technological University (GTU) *CGPA: 8.24 / 10*